

Alexander King, PhD

Computational Biologist & Data Scientist | AI/ML Systems & LLM Evaluation

Remote (based in Riverside, CA)

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PROFILE

Computational biologist and data scientist who builds end-to-end systems that turn messy, high-dimensional data into rigorous, reproducible results, and ships them. First-author in the *Journal of Neuroscience* on work I drove from computational discovery through bench validation, and solo-builder of shipped AI and evaluation systems: an LLM score-predictor at $R^2 = 0.94$, a 525-episode automated news pipeline, released statistical-audit software, and a study under review at TMLR. Full write-ups at alexanderjulianking.github.io.

COMPUTATIONAL BIOLOGY & GENOMICS

University of California, Riverside

2019–2026

Ph.D., Neuroscience (Computational Focus)

Riverside, CA

- First-author *Journal of Neuroscience* paper: discovered cell-type-specific alternative splicing of *Adgrl2* in single-cell RNA-seq (100K+ cells), then validated it at the bench with mouse genetics, blood-brain-barrier assays, and confocal imaging.
- Reproducible Python and R pipelines for single-cell and bulk RNA-seq on HPC/SLURM; integrated 6+ public GEO datasets. Trained 5 researchers; 4 years teaching, sole instructor for an upper-division course.

AI & ML SYSTEMS

Project Soothsayer

Solo-built

LLM evaluation & score-prediction system

[GitHub](#)

- Predicts Arena human-preference ELO from benchmark data alone (138 models, ~40% missing) with custom imputation and adaptive local regression. $R^2 = 0.942$, RMSE 13.90, $\rho = 0.966$ (5-fold CV; 0.937 walk-forward).

Newscaster

Live since 2024

Automated daily AI-news podcast, 525+ episodes

[YouTube](#) | [GitHub](#)

- Fully automated daily pipeline: a from-scratch multi-provider LLM router, an agentic source-grounded research loop (LangGraph), a hand-built RAG memory (**recall@10** $\approx$ **0.91**), and a self-correcting pre-TTS fact-checker.

EVALUATION & STATISTICAL AUDITING

Silent Precondition Traps for Data-Science Agents

2026

Independent research study, under review at TMLR (2026)

[Zenodo](#) | [GitHub](#)

- Found and reproduced a class of agent failures: an agent inspects a task thoroughly yet returns a confident wrong answer when a familiar tool silently assumes a precondition that fails in an uninspected slice of the data. Two frozen 10-task batteries, 1,000 runs: **26 to 100% failure rates**, 329 of 330 wrong runs hitting the exact planted error.

SC-Referee

2026

Statistical audit tool for single-cell analyses (Anthropic Life Sciences hackathon)

[GitHub](#)

- Released Python tool that audits single-cell analyses for statistical failure modes with abstention-aware, evidence-linked verdicts. On a published melanoma dataset, **16,289 “significant” genes fell to 770** (95.3%) when rebuilt at the biological-sample level.

PUBLICATIONS

- **King, A.**, et al. (2026). Endothelial *Adgrl2* expression and alternative splicing controls the cerebrovasculature. *Journal of Neuroscience*. (first author)
- Bern, M.^{*}, **King, A.**^{*}, Applewhite, D.A., & Ritz, A. (2019). Network-based prediction of polygenic disease genes involved in cell motility. *BMC Bioinformatics*. (*co-first)

SKILLS & EDUCATION

LANGUAGES & TOOLS Python, R, SQL, Bash; Git, pytest, HPC/SLURM, Google Cloud Platform, API integration

ML & STATS scikit-learn, pandas, NumPy; regression, Bayesian methods, imputation, PLS/PCA/SVD

AI & LLM Claude/Gemini APIs, RAG & embeddings, LangGraph agents, LLM-as-judge, benchmark design

BIOINFORMATICS single-cell & bulk RNA-seq, Seurat, Scanpy, Bioconductor, differential expression, GEO

EDUCATION Ph.D. Neuroscience, UC Riverside (2019–2026); B.S. Neuroscience, Reed College (2015–2019)